8.1 Measures of Center and Variation

Name _____ Period ____ Date _____

Algebra I 8.1 · Student sheet

I can…

- find and interpret the mean, median, and mode of a data set.
- explain how an outlier affects the measures of center.
- find the range and describe how changing every value changes the measures.

Vocabulary · fill in the missing word

Measure of center — one number that _____ the center of a data set: the mean, median, or mode

Mean — the sum of the values _____ by the number of values

Median — the _____ value when the values are in order (the mean of the two middle values when there is an even number)

Mode — the value or values that _____ most often (there may be no mode)

Outlier — a value that is much _____ or much less than the other values

Measure of variation — a number that _____ the spread of a data set

Range — _____ value − least value

Data transformation — adding the same number to, or _____ by the same number, every value

Part 1 · What's typical?

Hours of sleep last night for 10 students: 6, 7, 7, 8, 5, 7, 9, 6, 8, 7

Then an 11th student who slept **18 hours** joins the data.

Data	Mean	Median	Mode	Range
10 students				
With the 18 added				

Which measure changed the most? Which number best describes a typical student now? Why?

Part 2 · Notes: fill in the blanks

Mean = _____ ÷ _____ **Median** = the _____ value when the data are in _____

Mode = the value that occurs _____ **Range** = _____ − _____

An outlier pulls the _____ toward it. With an outlier, the _____ usually describes the center better.

Add k to every value: the center goes up by _____ and the range _____. Multiply by k: every measure is _____.

Part 3 · Examples (work them with the class)

Example 1

Quiz scores: 82, 90, 75, 90, 88, 79. Find the mean, median, and mode.

Example 2

The mean of 12, 15, 9, x, and 14 is 13. Find x.

Example 3

Commute times (minutes): 24, 27, 25, 26, 23, 25, 60.

a. Find the mean and median with the outlier.

b. Find the mean and median without the outlier.

c. Which measure better describes a typical commute?

Example 4

Points scored in 6 games:

Player A: 14, 20, 18, 22, 16, 18

Player B: 8, 30, 12, 26, 20, 12

Find the mean and range for each player. Compare.

Example 5

The data 4, 7, 7, 10 have mean 7, median 7, mode 7, and range 6. Find the mean, median, mode, and range when

a. 5 is added to each value. b. each value is multiplied by 2.

Part 4 · Partner practice

Partner 1

The mean of 7 test scores is 81. Six of the scores are 78, 85, 80, 74, 90, and 79. Find the seventh score.

Partner 2

Test scores: 70, 75, 82, 85, 88. The teacher adds 4 points to every score. Find the new mean, median, and range.